Journal of Energy Storage 25 (2019) 100844

Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est

# Vanadium redox flow batteries: A comprehensive review

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ARTICLE INFO

*Keywords:*
Renewable energy
Energy storage
Vanadium redox flow battery
Principles of vanadium redox flow battery
Design considerations of vanadium redox flow battery
Limitations of each component

ABSTRACT

Interest in the advancement of energy storage methods have risen as energy production trends toward renewable energy sources. Vanadium redox flow batteries (VRFB) are one of the emerging energy storage techniques being developed with the purpose of effectively storing renewable energy. There are currently a limited number of papers published addressing the design considerations of the VRFB, the limitations of each component and what has been/is being done to address said limitations. This review briefly discusses the current need and state of renewable energy production, the fundamental principles behind the VRFB, how it works and the technology restraints. The working principles of each component are highlighted and what design aspects/cues are to be considered when building a VRFB. The limiting determinants of some components are investigated along with the past/current research to address these limitations. Finally, critical research areas are highlighted along with future development recommendations.

## 1. Introduction

Climate change has proven to be one of the very consequential and critical environmental issues of recent time. Strong links have been drawn between greenhouse gases and the detrimental effects they have on the environment and climate change [1]. In the United States, most of the human produced greenhouse gases are the result of burning fossil fuels [2]. Approximately 78% of the generated power is produced from fossil fuels and roughly 13% of power is generated from renewable resources [3]. With the world's population historically rising by approximately 0.9% per year on average [4], future energy demands will concurrently increase. This rising issue creates a need for a solution. Shifting the power production practices to generate power from renewable resources will significantly cut down greenhouse gas emissions and in turn aid our environment.

In order to properly introduce renewable energy power generation sources, operation and management methods must be applied with intent to improve power supply quality and reliability [5]. Renewable energy sources rarely produce consistent, immediate power delivery in accordance to the ever-fluctuating power grid demand [6]. Converting renewable energy power to a storable source for subsequent deployment to the grid in a controlled manner is currently the favored method to accommodate this issue [6]. In designing a superior storage technique, the following characteristics should be considered:

1 **Scalability/Power Bridging** - It is important for the energy storage method to be scalable for large scale generation methods (above 100 MW) for discharge times over multiple hours and up to days [7]. Large scale operations will require these circumstances in order to properly manage the power generation.

2 **Fast Response Time** - To effectively execute power quality duties, fast response times are essential to mitigate voltage drops that occur during power generation [7]. Possessing fast discharge capabilities open up options, making the power storage type more versatile and pertinent to more applications [7].

3 **High Storage Capacity** - The ability to store power for prolonged periods of time will create maximum usability of the energy source. Most energy storage methods will slowly discharge over the duration of the storage period (through chemical losses in batteries, frictional losses in flywheels, etc.) and the overall efficiency of the energy cycle is lost along with power usability/versatility.

4 **Low Cost** - The cost of a storage technique often correlates with the technical maturity. As more manufacturers produce the product/components, the more competitive the market is and the lower the cost becomes. It is high priority for energy storage techniques to be inexpensive in order to drive down the cost of renewable energy initiatives in a market where non-renewable energy production is less expensive.

5 **High Efficiency** - The efficiency of a storage source is characterized by the rated output power divided by the rated input power. Amounts of energy are generally lost in the charging/discharging process, through self-discharge, friction, heat loss or chemical losses.

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https://doi.org/10.1016/j.est.2019.100844
Received 10 April 2019; Received in revised form 20 June 2019; Accepted 17 July 2019
Available online 31 August 2019
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**Table 1**
Energy storage method advantages and disadvantages.

| Storage Technique | | Advantages/Disadvantages |
|---|---|---|
| Sensible Thermal Energy Storage | Adv. | • Potential for low storage costs per kWh [10] |
| | Dis. | • Little development in power grid energy storage applications [11]<br>• High freezing points, increased susceptibility to energy loss [12]<br>• Low energy density [13]<br>• Short storage times due to the self-discharging phenomenon [13] |
| Latent Thermal Energy Storage | Adv. | • High energy density [14]<br>• Low weight/volume of material relative to sensible thermal systems [14] |
| | Dis. | • Low thermal conductivity (low charge and discharge times) [14]<br>• Limited operating temperature range [14]<br>• Low long-term durability from thermal cycling and corrosion (specifically phase change materials) [15]<br>• Little development in power grid energy storage applications [11] |
| Fuel Cell | Adv. | • High energy density, storage capacity [16]<br>• Highly scalable from small applications to large [17]<br>• No dependence on geographic locations<br>• Fast discharging times<br>• Hydrogen is a flexible medium used with other energy production applications [18]<br>• Modular system (ability to make unit larger at later time) [17]<br>• Nearly zero self-discharge (depending on the system) [16] |
| | Dis. | • Uses expensive hardware [16]<br>• Low systems life expectancy [19]<br>• Expensive production per unit energy (due to fuel/oxidant cost) [17]<br>• Low overall efficiency [16,19] |
| Lead-Acid Battery | Adv. | • Batteries generally economical, compact, easy to integrate with minimal effect on neighboring geology/architecture [20]<br>• Long useful life with high number of cycles [20]<br>• High recycling potential [20]<br>• Quick response time [20]<br>• Mature technology<br>• High sustainability relative to other battery types [20]<br>• Better safety than other battery types [21] |
| | Dis. | • Permanent damage if batteries not fully charged after each generation cycle - negative plate stops accepting charge (known as sulfation) [20]<br>• Short useable life before replacement or refurbishment (approximately 1200–1800 cycles) [16,19]<br>• Poor performance at low temperatures; requires thermal regulating systems in cool climates [17]<br>• Slower response times; lower efficiency<br>• Low specific energy (~30–50Wh/kg) [17]<br>• Risk of thermal runaway due to accelerated internal heat production resulting in failure and/or explosion [22] |
| Lithium-Ion Battery | Adv. | • Higher energy density then lead-acid battery storage [20]<br>• Mature and proven technology<br>• Fast response time [17]<br>• Minimal impact on surrounding placement geology |
| | Dis. | • Electrolyte is highly flammable<br>• Requires multiple fuses, thermal sensors and voltage monitoring for safe operation and avoiding thermal runaway [20,22]<br>• Harmful when fully discharged [16]<br>• Expensive energy storage because of the overcharging safeguard systems, manufacturing costs and materials (> $600/kWh) [19]<br>• Sensitive temperature and voltage operating range [16]<br>• Lithium is nonrenewable; overconsumption an environmental risk [23] |
| Capacitor/Super-capacitor | Adv. | • Very fast charging time<br>• Charge/discharge thousands of times with minimal loss and little to no maintenance [16,17]<br>• Very fast response time [24] |
| | Dis. | • Low energy density – any sizeable power requirement requires massive capacitors [16,17]<br>• High energy dissipation due to self-discharge loss [17]<br>• Large scale capacitors (energy densities over 20 kW h/m$^3$) are still being developed [17]<br>• Advanced materials have high costs to develop and use [16,25]<br>• Require cooling systems to operate effectively [16] |
| Flywheel Energy Storage | Adv. | • Long lifetime, short storage periods (usually around 10 s) with an approximate hold time of 10 min [26]<br>• Fully charge and discharge, no depletion to the amount of power expended over time [17,16]<br>• Highly efficient (~90%) [16]<br>• Wide temperature operating range [16] |
| | Dis. | • High speeds require significant maintenance such as bearing replacement to reduce frictional losses<br>• Short energy storage times and generation service lengths unsuitable for some renewable energy generation types [27]<br>• Mechanical failure risks substantial damage and poses a safety risk |
| Compressed Air | Adv. | • Long storage capabilities (greater than 10 plus hours) and relatively low capital costs [28]<br>• Mature technology; used for decades<br>• Inexpensive method to store power long-term [29]<br>• Low storage cost [16] |
| | Dis. | • Limited to where they can be built [30]<br>• Little flexibility for fast startup/ramp-up when power needed [29]<br>• Must coexist with gas turbine plants and associated combustion emissions [30,28]<br>• Isochoric plants (constant volume storage) undergo variable pressure during the charging cycle reducing efficiency [29]<br>• No "off the shelf" equipment available for highly efficient compressed air energy storage system [29]<br>• High uncertainty when choosing locations along with geographical restrictions unique to each location [29] |
| Pumped Hydro | Adv. | • Low cost energy storage method [31,10]<br>• Well-developed storage method [31]<br>• Long infrastructure life [32] |

(*continued on next page*)

**Table 1** (*continued*)

| Storage Technique | | Advantages/Disadvantages |
|---|---|---|
| | Dis. | • Require specific build sites [10]<br>• High capital costs [10]<br>• Environmental concern with the large infrastructure [10,32]<br>• Low energy and power density [10] |

Higher efficiencies ensures more of the produced power is transferred to the grid.

6 **Long Life** - Long operational life influences the cost factor, as lower life times will increase maintenance activity costs and operating costs. Downtime to replace failed components with the potential requirement for offline repair will also reduce the reliability of the storage system.

7 **Low Environmental Repercussions** - Ideally, the energy storage method will have little influence on the environment. The products used in the energy storage method should be sustainable, easily available or easily recyclable [8]. The environmental aspect does not merely end at the design and construction stages of the source, but in fact the analysis is completed for the entire life of the energy storage method. Maintenance, recyclability and any other factors must be considered to fully capture a storage methods "greenness" [9].

There are numerous power storage techniques being employed today. Various mechanical, electrical, chemical and thermal energy storage techniques have been developed in hopes to produce an optimal power storage method. Table 1 lists common and emerging energy storage methods and the associated advantages and disadvantages of each.

At present, the most popular large scale (> 100 MW) renewable energy storage technique is pumped hydro energy storage (PHES) [31]. Over 95% of energy storage capacity worldwide is currently PHES, making it by far the largest and most favored energy storage technique. This storage technique is mature and has been in use and applied at a large scale for many years. Benefits to this technology is the long energy storage times in relation to the alternate energy storage systems. The price per unit energy is comparatively low with modest operational and maintenance costs due to the simplicity of the system [31]. This is a system that is capable of start-up/operation during blackout periods and the infrastructure has a long operational life (expected to last between 50–100 years) [32].

Although the PHES technology has paired well with energy storage in the preceding years, some of the existing PHES facilities were built to perform large baseload generation and not the irregular, sporadic generation demanded by renewable resource power generation [32]. Large amounts of land are required to build the sizeable infrastructure associated with PHES. PHES systems pose environmental issues and high construction costs amidst the large infrastructure footprint and technical construction [31]. These systems also require specific plots of land in order to function; they need access to large bodies of water, access to the grid and large elevation disparities in order to house the reservoirs [32].

With the number of commercially available energy storage systems, there is no method currently available that fulfils all exemplary traits of an optimal energy storage system [7]. Emerging storage techniques such as the redox flow battery (RFB) hope to achieve these requirements. A key advantage to redox flow batteries is the independence of energy capacity and power generation. The capacity of the battery is related to the amount of stored electrolyte in the battery system, concentration of active species, the voltage of each cell and the number of stacks present in the battery [33]. Conversely, the power generated is related to the behavior of the active species and the electrode size [33]. This facilitates the vast scalability and flexibility of the technology. These are relatively simple systems with few moving parts and often require little operator input, making them low maintenance with little attention once set up and running [34]. The simple design nature also includes ease and possibility for modular construction [35]. The simplicity of the redox flow battery and the reversible redox reaction along with the presence of two soluble redox couples (removing solid-state reactions) can facilitate batteries that in theory, have no cycle life limit [36,37]. In some applications, these batteries can fully charge and discharge without substantial negative effects on the cell's components [36]. These systems can adapt well to a variety of power applications due to their prosperous short response times, on the order of milliseconds [38], indicating good fit for leveling power output from intermittent renewables. All of these advantages make the flow battery a very encouraging, important energy storage source for the future. The combination of all these properties allow the battery to have relatively low running and capital costs, especially compared to other emerging energy storage technologies [39].

On the contrary, RFBs generally have low energy densities, making them a non-favorable option for some applications where size and weight will affect the usability (ex. vehicle transportation, portable devices, etc.). Some redox flow battery implementations involve electrolyte reactions that create deposits of solid species within the cell [40]. These undesired side reactions reduce power density, capacity and overall efficiency [40]. The membranes used in these systems are very costly and cheaper alternatives are in constant pursuit. This will further be explored in the Membrane 2.3 section. As well, redox flow batteries are subject to additional parasitic losses along with the typical self-discharging losses; these unique losses stem from the pump work required to transport the electrolyte between the storage tanks and cells, and the electrical leakage due to shunt currents within the cell [38]. These losses can consume between 3–5% of the energy stored in the battery and must be kept minimal to maintain the efficiency of the battery [38]. Reducing the pump losses by increasing the cross-sectional area of the electrolyte channels, however, increases the leakage currents in the stack [38]. Further losses are sometimes necessary in order to maintain a constant electrolyte temperature due to its instability; this temperature control can cause a significant additional amount of energy loss [41].

There are several advantages and disadvantages associated with VRFB technology specifically. Some advantages of VRFB technology include:

- No solution contamination with diffusion of vanadium ions across the membrane [37,45,46]
- Regeneration of ion crossover occurs simply through normal battery operation [37,45]
- Moderate efficiency (between 70–90% for controlled laboratory setups) depending on the treatments and conditions exposed to the components [40]
- Lower levels of gas evolution during quick charge cycles relative to other flow batteries [37,45]
- Potential for electrolyte recycling between applications [45]

Conversely, some disadvantages of VRFB technology include:

- Thermal regulation systems to keep the system between 10 °C and 40 °C to avoid thermal precipitation of vanadium species [35,47]

- High oxidation properties of $V^{5+}$ deteriorate some ion exchange membranes and the positive electrode terminal [48]
- Gas evolution (produced at the electrode) can damage and lower cell efficiency [49–51]
- Battery charge depleted and electrode surface area reduced from gas evolution [50,52,53]

Given the benefits of VRFB technology, this review sets out to cover the current state of VRFB technology. Firstly, the design considerations of a VRFB and the associated working principles of the technology will be covered. The literature review section of this paper has been divided up in order to explore each component within a VRFB. For some components; design considerations, design aspects/cues, limitations and what has been/is being done to address each limitation was discussed. Current and past research efforts on some components are also discussed to highlight the work performed/being performed to further advance the technology. Finally, critical research areas are highlighted along with future development recommendations.

## 2. Literature review

The primary components of a VRFB include an electrolyte, membrane, electrode, bipolar plate, gasket, collector plate, storage tank and pumps. A literature review for these components was performed to further understand the design considerations, limiting factors and research to address the limitations.

### 2.1. Vanadium redox flow battery working principle

The most promising, commonly researched and pursued RFB technology is the vanadium redox flow battery (VRFB) [35]. One main difference between redox flow batteries and more typical electrochemical batteries is the method of electrolyte storage: flow batteries store the electrolytes in external tanks away from the battery center [42]. VRFBs generally use two such tanks to store vanadium ions in four different oxidation states ($V^{2+}$, $V^{3+}$, $VO^{2+}$ [otherwise $V^{4+}$] and $VO_2{}^+$ [otherwise $V^{5+}$]) such that there is a separate redox couple in each tank (see Fig. 1) [43]. Through the duration of this document, the positive side refers to the electrode and electrolyte which contains $V^{4+}$ or $V^{5+}$. The opposite applies for the negative electrolyte and electrode.

The electrolytes are fed through separate half-cells within the battery and are then returned to storage tanks for recirculation [42]. Each half-cell of the reactor is composed of an electrode and a bipolar plate; two half-cells are separated by a membrane to allow selective ion exchange while preventing the cross contamination of electrolytes [44]. This constitutes a single cell and adjacent cells share a bipolar plate to form a stack. The half-cell reactions for the positive and negative electrodes are depicted in Eqs. (1) and (2) respectively. The overall cell reaction is shown in Eq. (3).

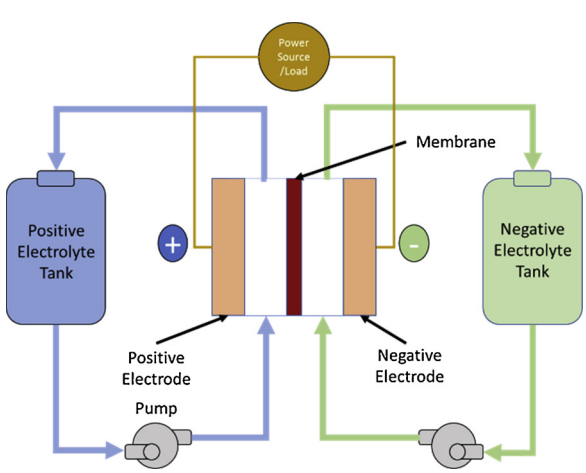


**Fig. 1.** Redox flow battery diagram.

$$\text{Positive cell reaction:} VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+} + H_2O \quad (1)$$

$$\text{Negative cell reaction:} V^{2+} \rightleftharpoons V^{3+} + e^- \quad (2)$$

$$\text{Overall cell reaction:} VO_2^+ + V^{2+} + 2H^+ \rightleftharpoons VO^{2+} + V^{3+} + H_2O \quad (3)$$

Multiple stacks can be electrically connected to create an array for larger applications [8]. Typical VRFB stacks and the cells within, are fed in parallel which maintains a constant concentration of redox ions in each stack/cell inlet, allowing for a more uniform flow rate and reduced overall pressure drop [8]. In a completely discharged state, the negative and positive electrolytes contain only $V^{3+}$ and $V^{4+}$ ($VO^{2+}$), respectively; during charging, the negative electrolyte is reduced to $V^{2+}$ while the positive electrolyte is oxidized to $V^{5+}$ ($VO_2{}^+$) [43]. Recharging the battery moves electrons through the bipolar plate from the positive side to the negative side and causes hydrogen ($H^+$) ions to diffuse across the membrane to the negative side [36]. The same reaction occurs in reverse during discharge. Early implementations of VRFB had energy efficiencies around 70% [37] and later developments led to efficiencies of over 80% [40].

### 2.2. Electrolyte

#### 2.2.1. Working principles: electrolyte

The two electrolytes in a flow battery react with each other to provide the electrical potential [54]. These electrolytes are comprised of an active redox species and a supporting electrolyte (solvent and supporting salt) [55] (Fig. 2).

In a VRFB, both electrolytes use the same active species, which reduces capacity loss due to cross-contamination of electrolytes [46], and generates an output voltage of 1.26 V [56]. VRFBs can typically store between 20 and 30 Wh/L of electrolyte, depending on the concentration [57]. Multiple vanadium compounds were first considered for use in various supporting electrolytes: vanadium trichloride ($VCl_3$), vanadium pentoxide ($V_2O_5$), and vanadyl sulphate ($VOSO_4$) were each considered with hydrochloric acid (HCl), sodium hydroxide (NaOH) and sulfuric acid ($H_2SO_4$) [45]. $VCl_3$ and HCl were excluded to prevent possible chlorine gas generation, and $V_2O_5$ was excluded due to its low solubility in the acids [45]. $VOSO_4$ and $H_2SO_4$ were thus initially selected [45], though shortly after, methods for creating electrolytes with $V_2O_5$ were developed [58,59]. As of now, research groups use $V_2O_5$ when aiming to reduce the cost of the electrolyte, and $VOSO_4$ when the vanadium concentration is intended to be easily altered [60].

#### 2.2.2. Cell design: electrolyte

*2.2.2.1. Preparation of electrolyte.* Initial preparation and charging of the electrolyte are crucial steps in the construction of an VRFB, some of the preparation methods are outlined in Table 2 below.

There are two major factors currently limiting the potential of the electrolyte for the VRFB application. Two aspect to consider during the design of a VRFB are:

**Specific Energy/ Energy Density –** The greatest challenge and largest area of research with the electrolyte is the low energy density, which typically limits VRFBs to stationary applications. The specific energy and energy density rely on the electrolyte's composition, and usually lie in the range of 15–25 W h/kg and 20–33 W h/L, respectively [64].

**Air Oxidation –** Another issue with the vanadium chemistry is air oxidation which more heavily affects the negative electrolyte but can easily be designed to avoid this [65]. $V^{2+}$ is readily oxidized to $V^{3+}$ by oxygen gas, which quickly discharges the battery. This effect is more drastic in solutions with a lower concentration of vanadium surrounded by excess oxygen and can be avoided by reducing the contact area between the electrolyte and air [65].

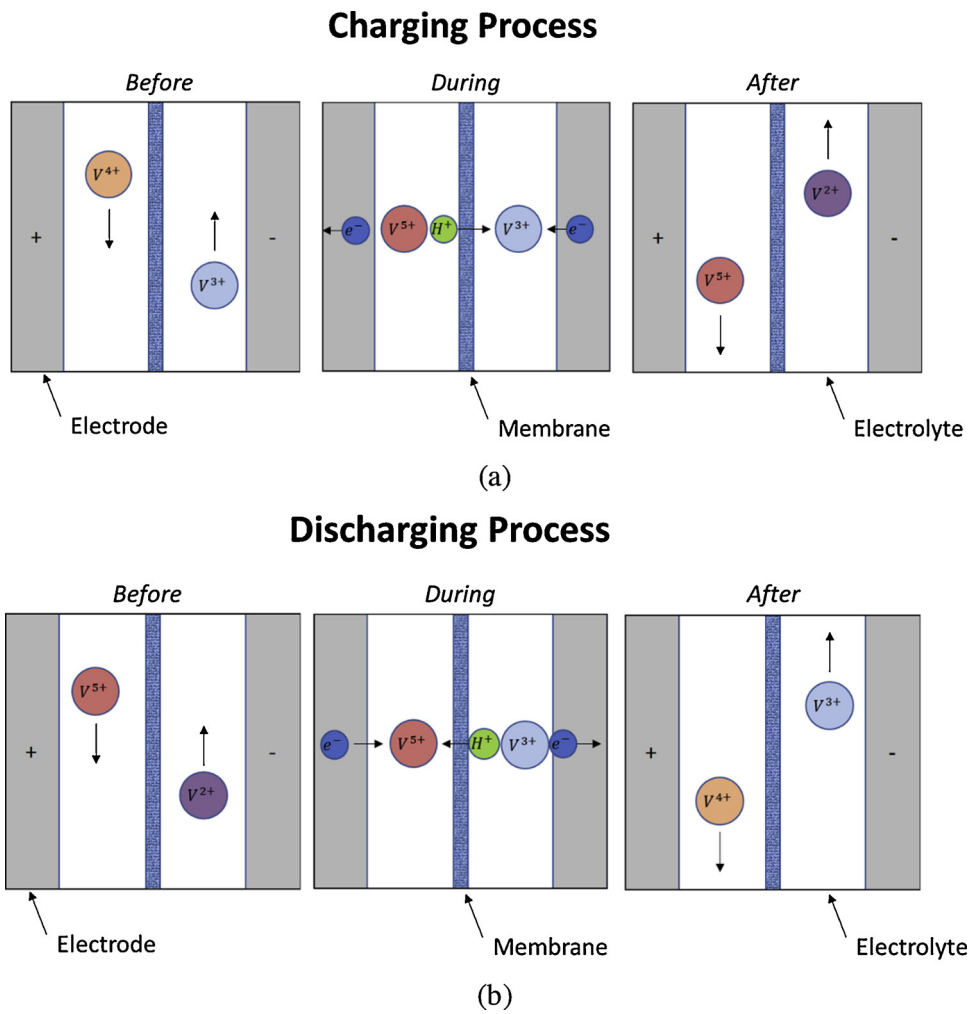


**Fig. 2.** (a) Charging and (b) discharging process depictions of a VRFB.

*2.2.3. Current research*

*2.2.3.1. Specific energy/ energy density*. Practical VRFBs rarely have vanadium concentrations exceeding 2 M [48]; this is limited by the solubility of the vanadium ions in the supporting electrolyte [66]. Kazacos et al. [47] showed experimentally that at temperatures above 40 °C, the vanadium and $H_2SO_4$ concentrations should be kept to 1.5 M and 3–4 M, respectively, since $V^{5+}$ tends to precipitate from $H_2SO_4$ when stored for long periods at high temperatures and states of charge. Similarly, $V^{2+}$, $V^{3+}$, and $V^{4+}$ have reduced solubilities in $H_2SO_4$ at temperatures below 5 °C [66]. At temperatures above 40 °C, $V^{5+}$ forms $V_2O_5$ [67], while $V^{2+}$, $V^{3+}$, and $V^{4+}$ form sulphates at temperatures below 10 °C [68] causing precipitation. The effects of temperature on the solubility of vanadium (specifically vanadyl sulfate) are shown in Table 3, which has been reproduced from Rahman et al. [69]. If the electrolyte temperature is consistently moderate and the battery is continuously cycled, the vanadium concentration can be increased to 3 M, which in turn can increase the energy density to 35 Wh/kg [66]. Further tests by Kazacos et al. [47] determined that precipitated $V^{5+}$ could be re-dissolved by mixing the precipitated positive electrolyte with negative electrolyte, or by discharging the electrolyte within the cell.

As a crucial component of the electrolyte, the concentration and composition of supporting electrolyte must also be considered; Mohamed et al. [70] showed a higher $H_2SO_4$ concentration improved the electrolyte's conductivity. Islam et al. [71] showed the increase also reduced the diffusion of water and vanadium ions across the membrane,

**Table 2**
Electrolyte Preparation Methods.

| Vanadium Compound | Reference | Charging Method | Description |
|---|---|---|---|
| $VOSO_4$ | [37,61] | Basic Charging | With three equal volumes of $V^{4+}$ solution, charge two parts in the positive side and one part in the negative side of an electrolysis cell to $V^{5+}$ and $V^{2+}$; dispose of one-part positive solution to get equal volumes of each electrolyte. OR Charge one part of negative and positive solution to $V^{3+}$ and $V^{5+}$, then replace the $V^{5+}$ with uncharged $V^{4+}$ and charge again to $V^{2+}$ and $V^{5+}$. |
| $VOSO_4$ | [60] | Waste Reducing | Charge one part of negative and positive solution to $V^{3+}$ and $V^{5+}$, then use a chemical agent (e.g. oxalic acid) to reduce $V^{5+}$ back to $V^{4+}$, then charge again to $V^{2+}$ and $V^{5+}$. |
| $V_2O_5$ | [62,61] | Chemical Reductant | Mix $V^{5+}$ solution with an agent like $V_2O_3$ or glycerol and perform a charging method like above. |
| $V_2O_5$ | [63] | Electrolytic Dissolution | With supporting electrolyte ($H_2SO_4$ and HCL) in both half cells of an H-cell, put $V_2O_5$ powder in the negative compartment, and electrolytically reduce to get a ratio of exactly 1 between $V^{4+}$ and $V^{3+}$. |

**Table 3**
Solubility of vanadyl sulfate ($VOSO_4$) in various sulfuric acid concentrations at temperatures from 10 to 50 °C. Retrieved from Rahman et al. [69].

| | | At equilibrium | | |
|---|---|---|---|---|
| Initial $H_2SO_4$ (mol/L) | Temp (°C) | Total sulfur[a] (mol/L) | $VOSO_4$[a] (mol/L) | KSIP (M) |
| 0 | 10 | 3.167 | 3.037 | 9.618 |
| 1 | 10 | 3.659 | 2.432 | 8.899 |
| 3 | 10 | 4.692 | 1.448 | 6.794 |
| 5 | 10 | 5.987 | 0.784 | 4.694 |
| 6 | 10 | 6.895 | 0.493 | 3.399 |
| 7 | 10 | 7.39 | 0.384 | 2.838 |
| 8 | 10 | 8.219 | 0.243 | 1.997 |
| 9 | 10 | 8.802 | 0.185 | 1.628 |
| 0 | 20 | 3.449 | 3.28 | 11.313 |
| 3 | 20 | 4.717 | 1.786 | 8.425 |
| 5 | 20 | 6.129 | 0.995 | 6.098 |
| 6 | 20 | 6.925 | 0.711 | 4.924 |
| 7 | 20 | 7.434 | 0.57 | 4.237 |
| 8 | 20 | 8.328 | 0.363 | 3.023 |
| 9 | 20 | 8.936 | 0.26 | 2.323 |
| 0 | 30 | 3.72 | 3.64 | 13.541 |
| 3 | 30 | 4.939 | 2.149 | 10.614 |
| 5 | 30 | 6.374 | 1.262 | 8.044 |
| 6 | 30 | 6.868 | 0.997 | 6.847 |
| 7 | 30 | 7.724 | 0.718 | 5.546 |
| 8 | 30 | 8.404 | 0.514 | 4.32 |
| 9 | 30 | 9.291 | 0.386 | 3.586 |
| 0 | 40 | 4.105 | 3.994 | 16.395 |
| 3 | 40 | 5.353 | 2.46 | 13.168 |
| 5 | 40 | 6.454 | 1.585 | 10.23 |
| 6 | 40 | 7.178 | 1.216 | 8.728 |
| 7 | 40 | 7.851 | 0.875 | 6.869 |
| 8 | 40 | 8.51 | 0.676 | 5.753 |
| 9 | 40 | 9.347 | 0.492 | 4.599 |
| 0 | 50 | 4.458 | 4.349 | 19.388 |
| 3 | 50 | 5.598 | 2.822 | 15.798 |
| 5 | 50 | 6.51 | 1.904 | 12.395 |
| 6 | 50 | 7.271 | 1.474 | 10.718 |
| 7 | 50 | 7.938 | 1.023 | 8.121 |
| 8 | 50 | 8.665 | 0.778 | 6.741 |
| 9 | 50 | 9.407 | 0.586 | 5.513 |

[a] Saturation concentration at equilibrium.

causing longer charging times, discharging times, and lower average voltages. Cheng [72] found that a decrease in sulfuric acid concentration increased the solubility of $V^{2+}$, $V^{3+}$ and $V^{4+}$, but decreased the solubility of $V^{5+}$. In 2014, Wu et al. [73] stated a 1.5–2 M $V^{5+}$ solution in 3 M $H_2SO_4$ was the optimal composition, and Mohamed et al. [70] found experimentally that 1.6 M $V^{3+}/V^{4+}$ in 4 M $H_2SO_4$ had the highest energy efficiency a year later.

Research has also been performed to characterize the VRFB's performance with solvents other than $H_2SO_4$. Peng et al. [74] tested a mixed acid solution of methanesulfonic acid ($CH_3SO_3H$) and $H_2SO_4$ and found a significant improvement in electrochemical activity over only sulfuric acid. The additional acid also increased redox reaction kinetics and decreased mass transport resistance. Chu et al. [75] showed that a similar mixed supporting electrolyte ($CH_3SO_3H$ and $H_2SO_4$) improved the solubility and stability of the vanadium ions with an increasing concentration of $CH_3SO_3H$, but this increase also negatively affected the solution's resistance and electrochemical kinetics. Roznyatovskaya et al. [76] compared performances of HCl electrolytes with $H_2SO_4$ electrolytes and found that, with electrolytes of the same conductivity and vanadium concentration, operation characteristics are similar between the two supporting electrolytes, but the cyclability of the $H_2SO_4$ cell was better. Further, the chloride ions in the positive HCl electrolyte were shown to parasitically reduce $V^{5+}$ at high SOCs (98%). Evidently, the composition of the supporting electrolyte is vital to improving the performance of the VRFB, and further optimization work is required in this area.

*2.2.3.2. Supersaturated solution*. A study to use a supersaturated positive electrolyte was performed by Rahman et al. [77], which showed the electrolyte is limited to 3.5 M $V^{5+}$ and 5–6 M sulfate/bisulfate since higher vanadium concentrations greatly increase the viscosity. The group determined a 3 M V, 6 M $H_2SO_4$ solution may suit the VRFB at moderate temperatures and cause a significant increase in energy density [77].

*2.2.3.3. Stabilizing additives*. Research has also been done on the effect of stabilizing additives to allow higher concentrations in the electrolyte solution, but the contrasting properties of the four vanadium oxidation states restrict this. Zhang et al. [67] tested $K_2SO_4$, which improved the stability of $V^{4+}$ but also increased the precipitation rate in $V^{5+}$. The same group showed certain additives like $Na_3PO_4$ stabilized $V^{3+}$, $V^{4+}$, and $V^{5+}$ exceptionally well in ex-situ tests, but caused inhibiting amounts of precipitation when tested within the cell [67]. It was hypothesized that the precipitation was catalyzed by graphite, showing care must be taken where other cell components are involved [67]. Wu et al. tested organic stabilizing compounds and found that inositol improved the reaction kinetics and the charge/discharge abilities and decreased the cyclic capacity fade in the positive electrolyte [78]. Chang et al. [79] determined 0.05–0.1 wt/weight of Coulter IIIA dispersant delayed precipitation in positive vanadium electrolyte at elevated temperatures. Polyacrylic acid was shown by Zhang et al. [67] to be an effective stabilizer for the negative electrolyte, and when mixed with methanesulfonic acid, it similarly stabilized the positive electrolyte in ex-situ tests. Wu et al. [73] tested additional acids in the supporting electrolyte hydrochloric acid to improve the stability and increase the stable temperature range, while methanesulfonic acid also stabilized by creating a complex to inhibit precipitation [73].

The stability of the electrolytes also varies greatly with the battery's state of charge [77]. Comparing electrolytes at 95% and 85% charge, Rahman et al. [77] showed that the two electrolytes exhibited 16% change and no change in concentration after 1000 h, respectively. An assumption by Moore [80] addresses the fact that the battery is ultimately unlikely to undergo a 100% discharge and will more likely cycle between 80% and 20%, validating the use of some higher concentrations.

*2.2.3.4. Novel chemistries*. The G2 vanadium redox flow battery developed by Skyllas-Kazacos et al. [64] (utilising a vanadium bromide solution in both half cells) showed nearly double the energy density of the original VRFB, which could extend the battery's use to larger mobile applications [64]. Skyllas-Kazacos [66] also discussed using polyhalides and complexing agents to avoid producing hazardous gases, which allows the battery to use $VCl_3/VCl_2$ in the negative electrolyte, and $Br^-/ClBr_2^-$ in the positive electrolyte.

Park et al. [81] improved VRFB chemistry by developing a composition of vanadium, manganese, and titanium in both the positive and negative electrolytes, where two ions react in each half cell. The group found that the optimal concentrations were 1.1 M of V, 1.5 M of Mn, and 1.5 M of Ti, with an energy density of 39.4 Wh/L [81]. Menictas et al. [82] created a hybrid vanadium-oxygen redox fuel cell with one electrolyte, reporting a specific energy above 40 Wh/kg. An et al. [83] proposed another fuel cell using a standard redox flow cell containing the vanadium chemistry with a separate reactor for chemically charging the cell with zinc and hydrogen peroxide.

## 2.3. Membrane

### 2.3.1. Working principles: membrane

The membrane is a component within the VRFB cell that keeps the positive and negative electrolyte solution from mixing [44,84]. This component is often referred to as a separator, and in general, there are two types that are applied to VRFBs: the ion exchange membrane and the nanofiltration membrane. While also preventing the cross mixing of

the electrolyte solution, the membrane must facilitate the redox reaction by allowing hydrogen ions to pass through, balancing and completing the net reaction [44,84].

### 2.3.2. Cell design: membrane

In flow battery applications, the membrane is crucial to maintaining a high efficiency over many cycles and the performance of the membrane greatly affects the net energy efficiency [84]. The largest obstacle the membrane component is facing is the trade-off between chemical stability and conductivity [84]. In past developments, alteration to the membrane to increase its resistance towards the VRFBs oxidizing environment decreases the ionic conductivity of the membrane and vice versa [44,84]. Currently, no perfect solution exists and the research and development towards membrane degradation in oxidizing environments is still underway [84].

Other factors that are imperative to the functionality of the cell are the membranes thickness and pore size. Increasing the pore size in the membrane will increase proton conductivity but also increase vanadium crossover [85]. This creates a balancing act in order to achieve a maximum efficiency. The thickness of the membrane is important as the thinner the membrane, the less flow resistance for the ion when passing through the membrane [85]. The membrane thickness must also be balanced, as a thinner membrane will reduce the membrane mechanical strength, making it susceptible to tearing during the compression/operation of the cell.

Everything considered, the two major properties that are currently limiting membranes performance are overall proton conductivity and ion selectivity.

**Overall proton conductivity** - This has a direct effect on performance and will decrease the voltage efficiency in the battery [85,86].

**Ion selectivity** - Low selectivity measures directly influences the overall coulombic efficiency [85,86]. No transfer of vanadium ions across the membrane will ensure maximum coulombic efficiency and any crossover of vanadium/other species into the opposing cell will result in self discharge and reduced energy efficiency in the cell [84].

When designing, modifying pre-existing, or selecting new membrane materials, it is imperative to scrutinize the materials conductivity and ion transfer performance to make an effective VRFB [85]. The large development fronts for the membranes includes ion selectivity, the proton conductivity and the membranes durability/stability. As mentioned previously, cross contamination largely affects the overall performance of the flow battery, as the vanadium crossover will react with the opposing vanadium species and will require regeneration [84]. In order to address the above considerations, numerous membranes have been developed. The two most common applied membranes are the Ion Exchange Membrane (IEM) and the Nanofiltration Membrane.

*2.3.2.1. Ion exchange membrane (IEM).* IEMs are cross-linked linear polymer chains, forming intricate 3D networks that come in forms of sheets, ribbons or tubes [44]. Ion exchange membranes (IEM) have ionic groups (either positive or negative) attached to the matrix structure of the membrane that are used as transporters during the exchange of the $H^+$ ion amidst the VRFB redox reaction [87,88]. Depending on whether the membrane is anionic or cationic, the membranes will allow its associated ionic group to pass through the membrane [87]. This property is leveraged by VRFB to allow $H^+$ ions to pass through the membrane in order to stabilize and facilitate the redox reaction.

There are a wide variety of characteristics to consider when selecting an IEM. Membranes are often classified by these characteristics, being their design (form), material, configuration, aperture size and driving force. In general, there are five membrane materials that have been applied to and researched in fuel cell and RFB applications: perfluorinated ionomers, partially fluorinated polymers, non-fluorinated hydrocarbons, non-fluorinated membranes with aromatic backbone, and acid-base blends.

*2.3.2.2. Nanofiltration membrane.* One of the emerging separator technologies is the nanofiltration membrane. Unlike the traditional ion exchange membrane, this membrane uses a pressure difference to drive the ion passage through the membrane composed of many small pores [89]. Nanofiltration membranes use basic geometry (pore size) as a permeation gate, in order to allow hydrogen ion passage and containment of all other species [88]. The radius of vanadium and its ions are much larger than that of $H^+$ so the pore size of the membrane is to be selected in order to allow passage of the hydrogen ions and not vanadium [88]. This gives nanofiltration membranes the advantage and potential to have very high ion selectivity; the closer the pore radius is to the hydrogen ion size, the higher the selectivity [88]. The pore size in nanofiltration membranes is crucial, affecting the overall performance [89]. Too small of pores will increase resistivity and too large of pores will promote cross contamination (and reduce energy efficiency).

Modification methods have been applied to nanofiltration membranes in pursuit to further performance. Developments performed by Zhang et al. [90] included a silica additive to the surface/pores of the membrane. Results were positive over a standard, untreated nanofiltration membrane. Improved ion selectivity, higher coulombic efficiency, energy efficiency and the resultant cost was still lower than the benchmark Nafion® membrane [90]. The overall energy efficiency with these results are ~80%, where traditional Nafion® membrane has reportedly achieved higher values [90].

### 2.3.3. Current membrane research

*2.3.3.1. Chemical stability.* The $V^{5+}$ ions in the positive half of the cell and electrolyte used in VRFB creates a harsh environment for the neighboring components [85]. As discussed in the electrolyte section, the bulk product in the electrolyte is sulfuric acid, which typically degrades materials and in this application, has the tendency to wear the membrane material [85,91]. In the positive half-cell, the $V^{5+}$ ion stability can be influenced by a number of factors such as the vanadium concentration, sulfuric acid concentration, temperature, pressure and additives in the cell [85]. A lot of newly developed membranes have not been tested for long term chemical stability unlike the well-established perfluorinated membrane. As reported by Schwenzer et al. [85] numerous studies released information attained over short testing trials, not over the extended timeframes that VRFB are intended to run for.

Research has been performed by Sukkar et al. [86] to further understand the causes and effect in the chemical stability subject matter. Membranes were soaked in $V^{5+}$ solutions, concentration levels of 0.1 M and 1.0 M in order to test their chemical stability and rate of oxidation [86]. The membrane exposed to 0.1 M of $V^{5+}$ showed to be less stable [86]. It was formulated that the swelling in the membrane in the 0.1 M solution led to the increase in pore diameter, allowing for the $V^{5+}$ ions to enter the membrane, where degradation can occur at faster rates [86]. Additionally, the ion exchange capacity proved to increase with increased soak time [86]. The chemical stability and degradation of membrane materials in VRFB applications is a lightly studied area. As recommended by *B. Schwenzer et al.* [85], further understanding and development is required to understand membrane deterioration, if the deterioration method is consistent across all membrane types/materials and its link between mechanical stress.

*2.3.3.2. Water transport.* As defined in the general reaction, transport of hydrogen across the membrane is required to stabilize and complete the reaction. There are two modes that drive the transportation process; vehicular transport and intermolecular transport (also known as the Grotthuss mechanism) [85,92,93]. Vehicular transport refers to the pairing and transit of the hydrogen proton to the water molecule [85]. The Grotthuss mechanism is the "hopping" of a hydrogen ion through permeation through the network of water molecules [85,92]. The Grotthuss mechanism has been found to be the more prominent transport method throughout the cell [85,92]. These mechanisms are

important to understand the overall functionality the membrane.

Water transport across the membrane transpires in several ways. Water crosses the membrane alongside proton transport (through the Grotthuss mechanism and vehicular transport) and alongside vanadium through the crossover process [85]. In cases where anion exchange membranes or non-ionic separators are present, water is transferred from the positive half-cell to the negative half-cell [85,94]. With applications of cationic membranes (the more commonly applied membrane in VRFB applications), the net water transport is from the negative half-cell to the positive half-cell [85,94]. This is important since the water concentration in a cell affects the overall battery capacity and energy efficiency [85]. In general, the net water transport across most commonly applied membranes is minor and overall negative impact is minimal [85].

*2.3.3.3. Membrane fouling.* Membrane fouling refers to the vanadium cation deposition at the surface of the membrane or deposition of the ions inside the flow channels of the membrane [95]. $V^{5+}$ ions have the inclination to deposit on the surface of the membrane due to the low diffusivity value of these cations [96,95]. $V^{4+}$ ions have shown to reside and obstruct the flow channels of the membrane [95]. Generally, perfluorinated membranes are composed of hydrophobic species, meaning they will not react with the hydrated electrolyte [95]. The sulfonic acid groups (located on the membrane pore channels) on the other hand have hydrophilic properties [95]. It was found that due to the low diffusivity of the vanadium ions, full dispersion into the flow channels cannot occur, rather they form contact ion pairs with the sulfonic acid group, and block the passageways [85,95]. These sulfonic acid groups are crucial for proton conduction [85,95]. Therefore, this bonding takes away net proton conductivity along with increasing flow restriction due to the confinement of the flow channel [85,95].

This membrane fouling condition can be alleviated through soaking the membrane in sulfuric acid [85]. Techniques to repel the cationic vanadium species have been applied to typical perfluorinated membranes, but results often affect and reduce the overall proton conductivity [85,97]. Electrodeposition treatment performed by Zeng et al. [97] is one of few methods that can be reduce the dampening effect of membrane fouling. This example of membrane treatment reduced overall water transport by ~3 times and $V^{4+}$ ion permeability by an approximate factor of 5 [97].

*2.3.3.4. Ion conductivity & ion selectivity.* As previously discussed, low ion conductivity and selectivity within a VRFB equates to an overall efficiency loss. Different membrane preparation methods can be applied to IEM to reduce this impediment. Traditional methods to create IEM include the polycondensation process [84]. The addition of special cationic or anionic groups to the IEM can change the overall ionic conductivity and selectivity of the membrane and improve performance [84]. Changes to the preparation process and the additives involved with the IEM have created the various types of IEMs as listed before (perfluorinated ionomers, partially fluorinated polymers, non-fluorinated hydrocarbons, non-fluorinated membranes with aromatic backbone and acid-base blends). Each IEM type has a specific preparation method, some including special additives. The motivation of exploring and creating the various IEMs is to see the effects of the component's ion conductivity and ion selectivity. All sub-IEMs possess different ion conductivity and selectivity characteristics and can be further explored in the work done by *Kariduraganavar et al.* [98].

### 2.4. Electrode

#### 2.4.1. Working principles: electrode

The electrode is the component that facilitates the oxidation and reduction reactions within the flow battery. The surface of the electrode acts as a catalyst for the reaction and its porous surface provides the reaction site for the electrolyte solution [99]. In most common implementations of a VRFB, there are two electrodes. One positive electrode (facilitating the reaction displayed in Eq. (1)) and one negative electrode (facilitating the reaction displayed in Eq. (2)).

#### 2.4.2. Cell design: electrodes

The reaction taking place at the electrode must be reversible and the oxidation and reduction species need to be soluble in the electrolyte to facilitate the reversible reaction. The states throughout the oxidation and reduction reactions must remain stable. The objective is to have the redox potential differences be as large as possible in order to maximize the electricity generated. For the flow battery to perform optimally, the electrode should have a high surface area and/or porosity in order to provide a large number of reaction sites and triple phase boundary [35]. The electrode material selection must be highly catalytic between the redox species in order to effectively react and charge the electrolyte flowing by and through it [35]. The highly acidic nature of the supporting electrolyte makes the cell environment very corrosive [35]. When sourcing an electrode material, it must have strong corrosion resistance. Since most electrolytes are sulfuric acid based, carbon and graphite materials are often used to form the electrode due to the materials corrosion resistance to sulfuric acid and good electrical conductivity [35,100].

When designing a VRFB, it is imperative to consider both electrode geometry and electrolyte flow [101]. These two factors will control the overall mass transport and therefore effectiveness of the battery. Before any systems are scaled up, developers must ensure the reaction surrounding the electrodes is controllable. Developers must know the relationship between the mass transport and the input flow rate for the set channel sizes and shape [101]. The surface area and current density of the electrode material should be maximized in order to provide a high current, all without jeopardizing any of the other components of the VRFB [101].

Two of the most common electrode materials used are graphitized polyacrylamide (PAN) and Rayon (cellulose fibers), in carbon paper and felt forms [102]. Carbon based materials purposed for electrodes have been a widely explored topic of research in previous years due to the materials diverse structure and advantageous chemical properties [103,34]. Electrode types can be further broken down into 2D and 3D electrodes: 2D electrodes being flat, smooth plates and 3D electrodes being more structured, porous materials [101]. Common three-dimensional, carbon-based electrodes include carbon cloth, carbon paper, and carbon felts. Advantages and disadvantages for all these forms vary. For example, carbon paper electrodes have a much lower electrical resistivity in comparison to the alternative 3D metals [103]. Other successes achieved with carbon paper electrodes include the very high-power densities attained (557 mW $cm^{-2}$), higher than that of any graphite felt [104]. Downsides to using carbon paper style electrodes include the limited reaction site availability and in turn, the limited amount of electrolyte flow through the cell. In order for the reaction to effectively occur and obtain a reasonable charge/discharge efficiency, a large amount of reaction area is required. With carbon paper electrodes, the flow through the cell must be lowered in order to compensate for this. More cells can be added to the stack, but the overall flow resistance will increase, and net efficiency will decrease due to the pumping power addition. If the number of cells and cost is restricted, the carbon paper electrode may not be feasible for the application due to this flow rate restriction. Some applications will require faster charge and discharge times that are dependent on the electrolyte flow rate through the cell, and the carbon paper may not facilitate this.

In comparison to typical carbon materials, graphite felt has a much higher electrical resistivity. Common carbon paper electrode materials have approximately 100 times lower electrical resistivity values [103]. Due to graphite felts natural hydrophobicity, various treatments have been developed to increase the accessible reactivity sites [103]. Graphite felt has been reported to perform strongly when suited as the negative electrode, but when used as the positive electrode, discharging

occurred during the charging of the battery cycle due to instabilities [37]. Mechanical and chemical disintegration can occur from this phenomenon after numerous cycles [103]. All these factors will influence and negatively impact the net energy efficiency of a VRFB.

After reviewing the literature, graphite felt electrodes appear to be the most common electrode material applied in VRFBs [34]. Graphite felt in VRFBs possess the advantage of bolstering local turbulence, improving cell mixing, thoroughness of reactions and mass transport throughout [103]. Graphite felt electrodes have the advantage of higher chemical conversion rate per electrolyte pass [105]. It is commonly the favored material for VRFB applications due to its chemical inertness, with small potential of reacting with the sulfuric acid-based electrolyte [103]. Although this is all typically true, untreated, natural felt electrodes have drawbacks in VRFB applications due to the poor wettability and the potential degradation under prolonged operation [106]. Additionally, due to the nature of the material, impurities will often exist. The impurities within the electrode will often create a net loss within the system due to the impurities reacting with the surrounding materials/species [103]. Graphite felt has had success in being applied in membraneless cell set-ups, proving to have high reactant conversion (> 90%) with relatively small net pressure drop [103,107]. Graphite felt has shown to have much lower kinetic polarization losses in comparison to configurations with graphite plates with single-serpentine flow fields due to its larger surface area [108].

The electrode is a crucial component within the VRFB and the overall performance of the VRFB is greatly affected by this element. Good design practices dictate that the losses within the electrode should be minimized as much as possible. Some losses that can be found in the electrode are listed in Table 4 [109]:

Through proper design or modification of existing electrode materials, the above losses can be minimized. Efforts have been undertaken through electrode property changes or material/structure modification in order to both fulfill the design cues and reduce the losses mentioned above. This will be discussed in more detail below.

### 2.4.3. Electrode current research

To improve the performance of the electrode, many different treatments and design aspects can be applied to increase the electrode electrochemical behavior, the number of reaction sites, and reduce its ohmic resistance. These treatments are highly researched and crucial to the topic.

*2.4.3.1. Carbon paper "zero-gap" architecture.* Recent research advancements using carbon paper/carbon cloth material electrodes have shown promising results. Carbon paper and carbon cloth paired with the "zero-gap" cell architecture by Liu et al. [104] have displayed valuable development. This "zero-gap" architecture refers to a configuration where the electrodes, current collector, and membrane are all in direct contact with each other [104]. The results were positive, with high power densities and very high current densities achieved [104]. Low surface activation (of the catalyst and the chemical reaction) is the current limiting factor with this design [101]. Researchers are actively seeking improvements to this through surface modifications to change a material's electrical conductivity [101].

*2.4.3.2. Heat treatment of electrodes.* Research has repeatedly shown that the inclusion of surface functional groups can impact the electroactive behavior of a VRFB [111,112]. Sun and Skyllas-Kazacos [112] set out to increase the oxygen functional groups of the electrode through thermal activation. With increased surface oxygen groups on the electrode, the better the electrode performed, and that curing the electrode at 400 °C yields optimal results [112]. Pezeshki et al. applied and verified these previously conceived theories of increasing the oxygen content on the surface of the electrode leads to performance gains throughout the cell [113]. This was verified by the varying of the oxygen content of the gas treatment, but this time on carbon paper. Overall, the heat treatment process in an environment with increased oxygen content improves the electrode performance and overall cell efficiency by 13% [113]. In terms of electrode treatment, heat treatment proves to be the most popular and practical (due to ease and lower cost) electrode treatment [102].

*2.4.3.3. Design enhancements.* One of the key methods to improve overall performance in the electrode is through design modification. One of the losses through the electrode is flow resistance, an aspect that researchers are continually trying to improve. Adding flow channels to the electrode can improve system efficiency by reducing the pumping power requirement, improve flow distribution over the reactive area, and lower flow resistance of electrolyte through the system [114]. *Bhattarai et al.* found overall efficiency increases of up to 2.7%, testing multiple flow path designs on the PAN based carbon felt [114]. These principles have been applied over carbon paper electrodes through integration of serpentine flow networks with a promising outcome [115]. As mentioned previously, mass transport within the electrode and VRFB is a limitation. One of the highly effective ways to curtail this loss is through the flow field design in the bipolar plate [34]. This will be further discussed in the bipolar plate section.

Other design advancements in the electrode component relates to new electrode material development. Carbon nanofibers, carbon nanotubes, and graphene oxide electrodes are all being developed currently [116]. Examples of applications and research of these materials include a graphene oxide composite electrode created by Li et al. [117], $TiO_2$ imbedded carbon nanofiber electrode created by He et al. [118] and the application of carbon nanotubes on the electrode by Li and Liu et al. [119]. All the mentioned applications help address some of the short comings of the traditional (graphite felt and carbon paper) electrodes, including better reactivity (and performance), and decreased electrode degradation.

Further performance improvements of these materials can be administered through the embedment of nanoparticles. $ZrO_2$ nanoparticles were embedded into the structure of a carbon nanofiber electrode by He et al. [120] with intentions of increasing the materials catalytic abilities. These nanoparticles were electrospun into the material due to the strong ability to cling particles that will not necessarily adhere/fasten well to others [120]. Results from this test were positive,

**Table 4**
Losses within the Electrode.

| Loss | Description |
|---|---|
| Kinetic Activation Polarization | The kinetic activation polarization losses are associated with the delayed charge transfer reaction occurring between the surface of the electrode and the surrounding electrolyte [109]. |
| Electrode Ohmic Losses | The electrode ohmic polarization losses are the resistances between the surface of the electrode and the electrolyte that reduce the effectiveness of transport of the ions [109]. The electrical resistance of the material and contact resistance between the electrode and the surrounding components can also contribute to the ohmic losses [109]. Electrode ohmic polarization losses are a heavily studied topic, and methods to reduce this effect are through the increased activation layers paired with high electrochemical activity on the electrodes surface [110,109]. |
| Mass Transfer Losses | Current produced from the reagent delivery to the electrode affects the net mass transfer and will impede the electrochemical activity. This can also be introduced from "starvation" to the electrode [109]. This can usually be adjusted through flow rate and pumping power [34]. |

showing that the $ZrO_2$ increased the components catalytic ability, increased the voltage and current efficiency by ~8% and showed signs of sustainability over 50 charge/discharge cycles [120].

*2.4.3.4. Compression.* The contact resistance between the electrode and current collector is greatly influenced by the compression in the cell [121]. On account of the geometry and construction process of VRFB cells, often the compression within the cell is nonuniform. Distributing the pressure evenly through the cell is often overlooked and can increase the minimum limiting current density with a lower net pressure drop [122]. The rib, channels, and component geometry all affect the deformation and distribution of compression throughout the cell [121]. The distribution of compression and performance effects were modeled and analyzed by Qu et al. [121], with the flow velocity, local current density, and overpotential distribution all varying from fluctuating compression values. In their conditions, the compression ratio of 55.7% provided the best results [121].

*2.4.3.5. Surface treatment.* Various surface treatments and methods have been applied to the different electrode materials in attempt to increase the overall performance. Some of the treatment methods are outlined in Table 5 below. One discovery that has helped the electrode in catalyzing the redox reaction (and overall wettability) is through the increase or inclusion of surface-active oxygen functional groups [123]. In practice, many researchers were able to do this through electrochemical oxidization, heat treatment, plasma treatment, and irradiation treatment [123].

*2.4.3.6. Chemical treatment.* Research efforts have continued to target and improve the electrochemical performance of the electrode through the use of chemical treatments [127]. Specifically, one way to treat the electrode is through administration of acid groups [127]. Early research done by Sun et. Skyllas-Kazacos [128] show the positive effects of treating VRFB graphite felt electrodes in a $H_2SO_4$ solution. Decrease in cell resistance with increased acid concentration treatment (up to a threshold limit around 98%) and increased performance in the cell were found in this experiment [128]. As previously reported, escalation of surface oxygen/functional groups is a historically known property that will increase the components' electrochemical performance [111,112]. This research further analyzed and quantified the importance of surface oxygen/functional groups and its relationship between performance gains [128].

Another example of chemical electrode treatment is the utilization of $H_2SO_4$ and $HNO_3$ acids through the ultrasonic mixing into hydroxylated carbon paper electrodes [129]. This process changes the surface oxygen/functional groups of the carbon paper from 3.8% to 14.3% [129]. This method proved promising, with increased VRFB cell efficiency and sustainability over 50 charge-discharge cycles [127].

## 2.5. Bipolar plate

### 2.5.1. Working principles: bipolar plate

The bipolar plate connects adjacent cells electrically and separates them hydraulically [130]. In practice, bipolar plates are often engraved with flow paths to help distribute the electrolyte through the half cell more thoroughly. These plates can have a wide range of conductivities (20–5000 S/m), which are directly related to the price of the plate. Noack et al. [131] found that a system exhibits similar costs when inexpensive, less conductive bipolar plates are used over fewer more expensive, more conductive bipolar plates for a constant power output.

### 2.5.2. Cell design: bipolar plate

Unlike fuel cells, VRFB bipolar plates cannot be metallic, due to the corrosive electrolyte. Typical VRFB bipolar plates are made of graphite because of its machinability, high conductivity and resistance to corrosion, though graphite plates also tend to be quite expensive to handle and difficult to assemble due to the material's brittle nature [130].

*2.5.2.1. Flow channels.* Utilizing flow channels can decrease the pressure drop across the cell and thus the required pumping power [132]. It is important to consider the channel design; a non-optimal flow path for the setting can increase the pressure drop across the cell, resulting in poor performance. Fig. 3 and Table 6 depict four common flow paths.

A trial by Kumar et al. [135] was performed between conventional, interdigitated, and serpentine and it was determined the serpentine path produced the lowest pressure drop. A previous study by Prasad et al. [136] for fuel cells showed contradicting results, as interdigitated flows had both a higher power density and lower pressure drop [135]. Fuel cell performance differs from redox flow batteries though. Further discussion indicates the dimensions of the bipolar plate, the number of channels and the permeability of the electrode all play a crucial role in the distribution of electrolyte, which could lead one path to be more effective than the other [135]. Since the path's performance is directly

**Table 5**
Electrode surface treatment studies.

| Treatment | Results |
|---|---|
| Nanodispersed particles of bismuth on the positive, graphite felt electrode along with a heat treatment of 450 °C of both electrodes in air [124]. | • Further stable, active reaction sites<br>• Demonstrated long term reliability<br>• Higher efficiency achieved |
| Coating of carbon nanoparticles was performed on the electrodes of a VRFB [108]. | • Increased net energy efficiency (~85%)<br>• Increased current density<br>• Improved electrocatalytic area on the electrode |
| Coating of NiO nanoparticles on the graphite felt electrodes of a VRFB [125]. | • Increased net energy efficiency (~20% gain over untreated graphite felt)<br>• Increased current density<br>• Improved electrocatalytic area on the electrode<br>• Durability over 300 cycles |
| ***CuPt***$_3$ nanoparticles secured to graphene substrate and nitrogen and oxygen (PAN) treated carbon electrodes [100]. | • Energy efficiency of ~84% achieved<br>• Increased current density<br>• Improved electrocatalytic area on the electrode<br>• Durability shown over 30 cycles |
| 3 different surface treatment methods tested in order to compare the effects/results of an untreated, mild oxidation at 500 °C for 5 h, oxygen plasma, and gamma ray treated graphite felt electrode [126]. | • The mild oxidation treatment was proven to have the highest voltage and energy efficiency<br>• The overall surface area was increased drastically when the mild oxidation was applied to the graphite felt, increasing 1344%<br>• Concluded that the performance of the vanadium ions heavily relies on the amount of reaction surface space supplied and the quantity and type of functional groups formed on the graphite felt are imperative in the influence on the kinetics of the redox reaction occurring |

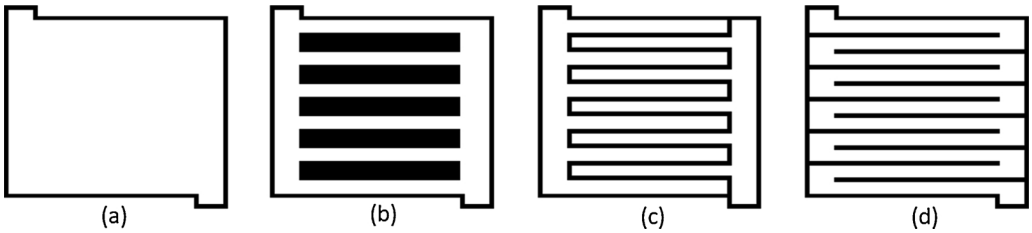


**Fig. 3.** (a) Conventional, (b) parallel, (c) interdigitated and (d) serpentine flow fields.

related to the specific conditions of the battery, it is difficult to state an optimal design. It is thus difficult to compare literature in this area due to varying configurations, though Table 7 compiles a selection of results additional to those mentioned above.

Flow fields have also been tested in the porous electrode. Of four designs considered by Bhattarai et al. [114], it was shown that the two interdigitated fields improved the use of the active surface. An analysis was performed by Reed et al. [133] which showed that the interdigitated channels cut into the electrode were the most effective, though the flow through design outperformed the other two designs at low flow rates and was slightly cheaper than the cut electrode.

#### 2.5.3. Bipolar plate current research

*2.5.3.1. Material development.* Using carbon composites could prevent the dissolution of graphite from fuel cell plates into the acidic electrolytes [130]. A study by Lee et al. [130] showed that the addition of 15 wt% carbon black exhibited comparable coulombic, voltage and energy efficiencies, despite a slightly lower electrical conductivity. A study by Park et al. [142] showed that a plate (denoted f-GKB-80) with flake-type natural graphite and ketjenblack nanoparticles displayed good conductivity (114 S/cm), structural properties and remained stable in the presence of acidic electrolyte. Han et al. [143] developed a bipolar plate without carbon by using $IrO_x$ coated $TiO_2$, which showed improvements in the voltage and energy efficiencies of 3–4% when compared to a standard graphite plate. Another composite utilizing carbon fabric/graphite hybrid was also shown to increase the energy efficiency by 6% when compared to typical graphite plates [144].

*2.5.3.2. Electrode flow.* There are three general methods for flow at the electrode: flow through, flow by, and flow across [8]. There are two typical meanings to 'flow through' which are not always specified in literature. A one-pass flow through allows the two electrolytes to mix after they pass through the cell stack [145], while a recirculating flow through maintains two separate electrolytes [146]. The former definition's reduced efficiency typically outweighs the economic advantage of omitting the membrane [145]. The second definition is quite simplistic, though still suffers from a large pressure drop at moderate flow rates [137]. One-pass flow through systems have extremely low optimal flow rates [145], which are often not feasible. Previous applications of the one-pass flow through configuration have had little success with high chemical conversions with the single pass nature of the arrangement [103].

### 2.6. Gasket

#### 2.6.1. Working principles: gasket

Gaskets are used in the VRFB application in order mate coupling components and create a seal between them so that electrolyte will not leak out. Often gaskets are placed between the bipolar plate and the membrane, separating the two so the electrode can make adequate contact with both components.

#### 2.6.2. Cell design: gasket

From the literature review (Table 8), the majority of the gaskets used in the VRFB application include PTFE, PVC and silicone materials. In VRFB cell design, the electrode will normally require some level of compression in order to achieve a satisfactory level of contact between the membrane and bipolar plate. The gasket design and overall gasket thickness can be leveraged in order achieve this optimal contact/pressure.

Harder materials (such as PTFE) are often used to create the required gap between the membrane and bipolar plate. For it to properly seal within the stack without using excessive amounts of pressure, a secondary sealant can be applied [152]. Applying sealant by hand (opposed to an automated, more precise process) can risk the deformation of the PTFE gasket/frame due to the uneven application and lead to cell leakage [152]. This is lesser of an issue with utilization of silicone style gaskets.

### 2.7. Collector plate

The term 'current collector' is often used indiscriminately in literature; some reports refer to copper plates at either end of the stack as current collectors [153] while others equate them to bipolar plates [152]. In a single cell assembly, the simple configuration allows the current collectors and bipolar plates to have the same use, so copper plates are often omitted. If the current collector is to be a metallic plate, the two criteria are a high conductivity and low cost, and if the metallic plates are omitted, the typical criteria for the current collector are the same as the bipolar plate.

If copper is used, damage to the bipolar plate can allow electrolyte to seep through to the metal, causing copper contamination, promoting hydrogen evolution in the negative half cell and failure of the battery stack [153]. Reynard et al. [153] studied this issue in a commercial VRFB and developed a process to plate out dissolved copper with the electrode, then remove it with an oxidative solution.

**Table 6**
Four common flow paths for VRFBs.

| Flow Path | Description |
|---|---|
| *Conventional* | A flow through configuration, where electrolyte is forced through the electrode. At higher flow rates, the increased hydraulic resistance causes a high pressure drop, though Reed et al. [133] showed that at lower flow rates this flow method performs well. |
| *Parallel* | Multiple parallel channels that connect to the cell's inlet and outlet via a header channel (Fig. 3a). |
| *Interdigitated* | Multiple parallel channels that are blocked so adjacent channels do not have both an inlet and an outlet (Fig. 3b). This design forces a hydraulic convection as the electrolyte passes through the electrode, and across the barrier separating adjacent channels. |
| *Serpentine* | A single channel that passes back and forth along the length of the cell (Fig. 3c). There are also serpentine designs containing a path that splits into multiple adjacent serpentine channels. The additional paths ultimately help to decrease the pressure drop across the plate [134]. |

**Table 7**
Comparison of four common flow paths for VRFBs.

| Paths Compared | Reference | Cell Specifications | Results |
|---|---|---|---|
| Parallel, Interdigitate, Conventional | [137] (Experimental) | • 23 $cm^2$ active area, 45 channels, 45 mm channel length, 1.17 mm × 0.76 mm channel dimensions<br>• 1.5 M V in 2.6 M $H_2SO_4$ as fluid<br>• 20% and 50% compressions of SGL Felt (Lower density, higher uncompressed thickness) and CP-A (higher density, lower uncompressed thickness) electrodes | • Higher flow rates caused higher pressure drops<br>• Conventional flow with SGL felt had the highest pressure drop (~8 kPa at 60 mL/min), parallel and interdigitated fields had comparably low pressure drops (~2 kPa at 60 mL/min) (the group states the latter is only semi-quantitative due to discrepancies between theoretical and experimental results)<br>• Interdigitated flow showed a current density over 1 A/$cm^2$ at 120 mL/min, while parallel flow showed less than 0.1 A/$cm^2$ at the same flow rate<br>• In a scale-up calculation, the conventional flow was shown to have the highest pressure drop above a 5 cm channel length, and the parallel flow field consistently maintained the lowest pressure drop at 60 mL/min |
| Conventiona, Serpentine, Parallel | [138] (Mathematical Model) | • 100 $cm^2$ active area, 3 mm thick electrode, 3 mm channel width and height | • Between the range of 5–20 mL/s, parallel flow field consistently showed the highest electrolyte concentration uniformity factor, while serpentine and conventional flows performed worse than the previous field, respectively<br>• The parallel field also displayed the lowest pressure drops while the conventional had the highest<br>• At low flow rates (5 mL/s), the conventional field had the lowest overpotential, though at higher rates (7–30 mL/s), serpentine flow rate performed better; the parallel field consistently showed the highest cell overpotential<br>• The serpentine flow field showed the highest energy efficiency and round-trip efficiency |
| Interdigitate, Serpentine | [139] (Experimental and mathematical) | • 9 $cm^2$ active area, carbon paper electrodes, Nafion™ 117 membranes<br>• 1.7 M $VOSO_4$ in 3.3 M $H_2SO_4$ as electrolyte<br>• Flow fields machined from BMC 940 graphite composite blocks | • Interdigitated flow fields were shown to have the lower pressure drops than serpentine both experimentally and mathematically at flow rates between 10 and 100 mL/min (at 100 mL/min the experimental data for interdigitated and serpentine were ~50 kPa and ~125 kPa, respectively)<br>• Interdigitated flow fields performed best when a low flow rate was used with a thin electrode, though still outperformed serpentine when a thicker electrode was used |
| Serpentine, Interdigitated | [140] (Experimental and mathematical) | • 25 $cm^2$ active area<br>• 1M V ions in 5 M $H_2SO_4$ is used as electrolyte | • Serpentine flow field showed better performance, but the pressure drops increased greatly with increasing flow rates (approx. 110 kPa at 60 mL/min, compared to 20 kPa with interdigitated field)<br>• Interdigitated performed better at high flow rates and low operating currents<br>• Interdigitated flow fields showed a more homogenous reaction rate with the lower velocity through the electrode |
| Variations of Interdigitate, Serpentine | [141] (mathematical) | 5 configurations tested:<br>1) 10 $cm^2$ serpentine<br>2) 50 $cm^2$, 4-channel serpentine<br>3) 50 $cm^2$ loose interdigitated<br>4) 50 $cm^2$ tight interdigitated<br>5) 400 $cm^2$ interdigitated<br>6) 0.4 mm, 1.2 mm, 2.4 mm electrode thicknesses | • Interdigitated field with tight channels had the lowest pressure drop compared to the specific flow rate for all electrode thicknesses, until flows increased much above 10 mL/min/$cm^2$<br>• The pressure drops were shown to scale non-linearly and were dependent on the cell area and flow conditions<br>• Areas of the cell which show experimental degradation correlate to areas with comparably low rates of transport |

**Table 8**
Gasket materials used in few research applications.

| Reference | Gasket Material | Information/results |
|---|---|---|
| [147] | PTFE Coated fiberglass | PTFE fiberglass sheet was used in this experiment, with the thickness of the gasket selected so that under total cell assembly, the electrode width would compress 20%. |
| [148] | PVC Gasket | Varying gasket sizes used to vary the compression of the electrode to see the different internal resistance. 0%, 10%, 20% and 30% compressions were tested. 30% compression had the highest capacity and power but 20% provided overall energy efficiency. |
| [149] | Not specified | Gasket was sized to compress the electrode approximately 80% of its natural thickness. |
| [150] | PTFE Gasket | Gasket was sized to compress the electrode approximately 75% of its natural thickness. |
| [135] | Silicone | Gasket was compressed 80% and carbon felt electrode was restricted to 0.19 compression ratio. |
| [151] | Silicone | Gasket compressed from a 3 mm to 2.1 mm thickness, 70% of its original size. |
| [152] | PTFE Gasket | Sealant required to form seal. |

### 2.8. System design

#### 2.8.1. Shunt currents

Shunt currents are losses from low resistance paths that bypass into the common manifold of the stack [154]. Since the electrolytes are good electric conductors, they can direct some of the produced electric current away from the external circuit which impacts performance and efficiency [155]. An ideal flow path on the bipolar plate is one that decreases shunt currents, while still reducing the pressure drop in the flow channels [156]. Models and studies performed by *Tang et al.* [154] have shown that the shunt currents in VRFBs can be reduced through construction of channels of longer lengths with smaller cross-sectional areas; though stronger pumping capabilities will be required [157]. This design factor will require compensation – the pump is often powered by the VRFB and optimization between the pump requirement and channel size/number should be adjusted to obtain an optimal net efficiency [157].

#### 2.8.2. Gas evolution

Gas evolution is a destructive phenomenon that can occur as a result of side reactions at the electrode from produced oxygen and hydrogen gas in the positive and negative electrodes, respectively [49–51]. These reactions expend electrolyte and alter the overall concentration of the active redox species. The reactions thus have an overall dampening effect on the battery and negatively affects all components of the cell. The hydrogen and oxygen evolution occur as a result of the following reactions, respectively: [51,52]

$$2H^+ + 2e^- \rightarrow H_2 \tag{4}$$

$$2H_2O \leftrightarrow O_2 + 4H^+ + 4e^- \tag{5}$$

These side reactions lower the cell's efficiency by removing some of the current density used to charge the cell [52] and also by reducing the useable surface area of the electrode [53]. More specifically, hydrogen evolution drains battery charge and reduces the reactive surface area of the negative electrode due to an accumulation of hydrogen bubbles on the electrode surface [50,51]. Hydrogen evolution is largely influenced by temperature: with high operating temperatures comes high rates of hydrogen evolution [50,51]; as well, increased current densities [51], and the size of the reactive surface area of the carbon electrode affected gas evolution on the negative side [158]. It was also found that hydrogen evolution occurred more rapidly at low states of battery charge [50]. This reaction amplifies the electrolyte charge imbalance caused by cross contamination, leading to further capacity loss over long cycling periods and reduces the number of available active sites on the porous electrode surface [53]. The occurrence of this reaction depends on the proton density and state of charge (SOC) of the battery [159]. It was hypothesized by Shah et al. [51] that increasing the flow rate of the electrolyte could reduce the production of gas. Contrarily, the positive electrode is more susceptible to oxygen corrosion [50]. The evolution of oxygen was found to increase at elevated operating temperatures [52].

Gas evolution also has a damaging effect on the bipolar plates. Satola et al. [160] tested bipolar plates ex-situ in the positive and negative electrolytes to see how gas evolution and overcharging would affect the plates. The group found that the hydrogen gas evolution at the negative electrode caused negligible damage to the bipolar plate after high discharge conditions, although a substantial amount of gas was produced [160]. On the positive side of the cell, the evolution of oxygen gas caused significant corrosion damage to the plate; the surface of the plate was oxidized, which was hypothesized to allow electrolyte to seep into the surface of the plate. At high overpotentials, the carbon was eroded, causing CO and $CO_2$ evolution, which expands the pore system creating larger cracks and further seepage [160]. Evidently, care must be taken to avoid overcharging, to avoid negative impacts of aging on the bipolar plate. Further, due to its effect on the entire VRFB system, gas evolution should be considered when designing and applying a battery.

#### 2.8.3. Pumps

Tang et al. [156] showed the importance of flow rate optimization for the efficiency of a flow battery by demonstrating the relation between overpotential, pump losses and the flow rate; the circulation also removes heat, reducing precipitation [156]. This study also showed the benefits of a variable flow [156]. An experiment performed by Ma et al. [161] showed similar results: increasing the flow rate near the end of both the charging and discharging cycles, accounts for the decreasing state of charge - this also improved the system efficiency. A mathematical and experimental study performed by Khazaeli et al. [162] showed a higher flow rate increased the cell voltage, and allowed the battery to remain operational particularly at low states of charge.

Further consideration must be made towards the flow direction of each electrolyte. In either equicurrent or countercurrent mode, the electrolyte inlets and outlets are on similar or opposing sides of the stack/cell, respectively [163]. The countercurrent mode exhibits improved operating conditions because each cell causes a similar drop in electrolyte charge which decreases the variance in voltage in adjacent cells and reduces the water transfer unbalance in each cell [164].

#### 2.8.4. Tank configuration

There are two possible tank configurations which allow for either multi-pass or single pass operations [163]. Recirculation mode constantly pumps electrolyte through the battery which only requires two tanks, while batch mode is a four-tank system where the electrolytes move from supply tanks to storage tanks and are returned after all of the electrolyte has drained through [165]. Batch mode provides a volumetric measure of the battery's state of charge [163], but recirculation mode is simpler, more efficient and more economical in construction and operation, and is typically the method used [165]. A three-tank system can be used, typically with a one-pass flow through configuration at the electrode, in which two supply tanks lead to a single storage tank for the mixed electrolyte, but this system is inefficient for the same reasons as a one-pass flow through model [166].

Ideally, the tank system within a VRFB will be sealed. There should be as little contact as possible with the electrolyte and any air. As discussed, $V^{2+}$ is eagerly oxidized to $V^{3+}$ by oxygen gas [65]. This is a system loss and effective design of a tank system will minimize this. Some methods to combat the oxidation is through the purging of the tank system with nitrogen gas [60,167,168]. The system is purged with the gas and then sealed before testing in order to minimize system losses [168].

*2.8.4.1. Tank size.* The reservoir tank size must be designed for the storage need and application. It is crucial that the storage system is not overdesigned, as this directly influences a system's impact on the environment. Oversizing the storage tank induces extra material usage, manufacturing/power expenditure and land/footprint acquisition [9]. A small-scale model by Arbabzadeh et al. displays the importance of system size optimization by calculating the total emissions created by off-grid energy systems [169]. Assuming a reduction of green energy would instead be levelled with natural gas combustion, the group showed that oversizing the battery reduces the environmental burden, despite the larger environmental burden associated with the manufacturing and material usage [169].

#### 2.8.5. System efficiency

The efficiency of a VRFB is dependent on many factors; the construction of the flow cells can drastically affect the system's efficiencies. In reported literature, it is not unusual to see moderate energy efficiencies [41,68,133,149], even in earlier demonstrations [37]. Holland-Cunz et al. [170] displayed an energy efficiency of 87% in an extended 20-day laboratory procedure, showing the possibility of extension to commercial use. However, there is little to no reported literature

discussing the system efficiencies of any commercially available VRFBs to date.

## 3. Conclusion

In this review, we have covered the working principles of common components involved in a VRFB system, theory, design queues and limiting factors within each component and current research to solve such limiting factors and/or to further understand the operating principles behind the component. From the multiple components reviewed, few configurations and design variants stand out as the leading VRFB application option. The following ideas can be drawn from this review;

- The zero-gap architecture has shown to be the leading cell configuration in terms of power density, overall efficiency and net cost.
- Zero-gap architecture can use either carbon paper or felt style electrodes. These two electrodes can perform similarly under the high compression involved with this architecture, with the trade-off being the durability (membrane punctures involved with the felt electrode and the high compression) and the lower surface area for the chemical reaction involved with the carbon paper electrodes.
- The electrolyte with the highest energy density was shown to be 1.6 $V^{3+}/V^{4+}$ in 4 M $H_2SO_4$, though increasing this is a major area of research and is likely to change in the near future.
- The current VRFB application favourite remains to be variations of the Nafion® membrane, and with the zero-gap cell architecture, the thinner the membrane that can be applied, the better the performance and efficiency.
- Various treatments and research to improve this component is still underway, as highlighted in Section 2.2.2 *Cell Design* and in Section 2.3.3 *Current Membrane Research*.
- The optimal bipolar plate and the flow field design depend on the application, component size, electrolyte composition and parameters run throughout the cell, as seen through Table 7.
- It has also been shown that a variable flow rate improves the battery's performance, especially near the end of the charging and discharging processes.

Although the VRFB is a hopeful energy storage method, there still exists some flaws and shortcomings. This includes:

- the technologies high cost,
- the thermal precipitation within the electrolyte,
- the limited energy density that can be achieved in the electrolyte and
- the degradation that can occur within the cell from the harsh environment created by the sulfuric acid and vanadium species.

Improvements to the system can come from improved electrolyte compositions, and more effective materials for the membrane, electrode, and bipolar plate. Reducing resistive and parasitic losses in the stack is crucial for maintaining high efficiencies and validating the use of VRFBs in commercial applications.

## Acknowledgement

This work is partially supported by Natural Sciences and Engineering Research Council of Canada through the Engage Grant Program and Virox Technologies Inc., an industrial research partner.

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